

Real-World Evidence Against Ground Truth

Validating a Causal Inference Pipeline on Semi-Synthetic EHR Data

Goal

Given confounded observational data comparing two drug classes on a clinical outcome, can standard propensity-score estimators recover the true causal effect, and how much can we trust that recovery?

Methodology

1. Built a realistic type 2 diabetes, metformin new-user cohort (1,551 patients) from Synthea synthetic EHR data
2. Injected a known treatment assignment (SGLT2i vs. DPP-4i) and outcome (heart failure hospitalization) via logistic and exponential-hazards models, embedding a disclosed confounding structure and a literature-anchored true hazard ratio of 0.67
3. Estimated the effect with naive, IPTW-weighted, and doubly-robust Cox models, scoring each directly against the known ground truth
4. Replicated the full pipeline across 500 Monte Carlo seeds to test calibration
5. Stress-tested the propensity model via a leave-one-out confounder-omission sweep, a negative control outcome, and an E-value, showing that omission severity depends on both a covariate's own effect size and whether a correlated proxy for it remains in the model

Key Insights

- IPTW recovers the injected effect closely under replication: mean HR 0.671 against a true 0.670, with 94.6% empirical confidence-interval coverage against a 95% nominal target
- The naive, unadjusted comparison is severely and systematically biased (mean HR 0.523, only 70.6% coverage), confirming the injected confounding is strong enough to mislead if left uncorrected
- The doubly-robust estimator looks plausible in the single run but fails under replication: it does not fit in 11.6% of seeds and returns an extreme, unusable hazard ratio in a further 3.6%, both traced to small-sample multicollinearity
- IPTW restores full covariate balance, all six covariates' |SMD| below 0.1 after weighting, down from a pre-weighting imbalance as large as 0.56 on the most confounded covariate
- Confounder-omission severity tracks correlated proxies and effect size together: dropping age is by far the most damaging (no correlated proxy available), while eGFR, the single largest injected effect, and creatinine ($r = -0.55$ with eGFR) partly buffer each other's omission
- The negative control outcome is correctly recovered by both naive (HR 0.803) and IPTW (HR 1.112), and the IPTW E-value of 2.20 sets a meaningful bar for the unmeasured confounding needed to explain the finding away

